A Search for Lorentz Invariance and CPT Violation at the NOνA Near Detector

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Abstract

Lorentz invariance and charge-parity-time (CPT) symmetry are fundamental assumptions of the Standard Model of particle physics. The Standard Model Extension (SME) is an effective field theory which predicts Lorentz invariance (LV) and CPT violation at the Plank scale ($\mathbf{10^{19}}$ GeV). We present a framework to constrain possible LV and CPT violating effects using data from the NOνA near detector. We model predicted LV effects by the first-order effective Hamiltonian as sidereal modulations in neutrino event rates. We analyze these event rates using a Fast Fourier Transform (FFT). A $\mathbf{3\sigma}$ detection threshold is established from the FFT power spectrum using a Rayleigh fit. Using this criterion, we set upper limits on nine SME coefficients governing the strength of LV, probing simulated LV-driven signals in the $\nu_\mu \rightarrow \nu_e$, $\nu_e \rightarrow \nu_\mu$, and $\nu_\mu \rightarrow \nu_\mu$ oscillation channels enabled by the enhanced ν_e statistics available at NOνA.

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1. Introduction

Lorentz invariance and CPT symmetry are foundational principles to all of physics. Lorentz invariance posits that the laws of physics remain unchanged between all inertial reference frames. CPT symmetry requires invariance under the combined operations of charge conjugation (C), parity inversion (P), and time reversal (T). Together, these symmetries underlie the quantum field theory and the Standard Model. However, theoretical quantum gravity formulations suggest that these symmetries could be violated at the Planck scale (10^{19} GeV).

Given how fundamental these symmetries are, we should test them. The Standard-Model Extension (SME) is an effective field theory that contains all possible Lorentz and CPT violating terms [4]. In the neutrino sector, these violations are described by an effective Hamiltonian, leading to directionally–dependent oscillation probabilities due to Earth's relative rotation in a Lorentz invariance violating background field. This leads to potentially observable modulations in the neutrino event rate as a function of local sidereal phase (LSP).

The NOνA near detector is effective for this search. The high quantity of ν_e events and precise timing allow us to examine short-baseline neutrino oscillations with sensitivity to direction, energy, and baseline dependent effects. Furthermore, the suppression of standard oscillation signal at the 1 km baseline enhances sensitivity to sidereal variations. This is the first dedicated search for Lorentz invariance violation at NOνA, expanding upon past studies by MINOS [2] and MiniBooNE [3] that primarily constrained SME coefficients governing the $\nu_\mu \to \nu_\mu$ and $\nu_\mu \to \nu_e$ transitions.

This technical document describes the framework to determine the upper limits for each LV violating coefficient predicted by the SME under the assumption that no statistically significant sidereal modulation is observed in the data. We will analyze simulated modulation in the $\nu_\mu \to \nu_e$, $\nu_e \to \nu_\mu$, and $\nu_\mu \to \nu_\mu$ channels. We convert spill timestamps to LSP using `AstroPy`, simulate neutrino events based on the measured protons-on-target (POT) distribution, then normalize event rates to remove the effects of time-dependent beam exposure variations. Using the Fast Fourier Transform (FFT), we determine the power spectrum of the first four sidereal harmonics. A Rayleigh fit to the FFT power spectrum in simulated unmodulated experiments establishes a 3σ detection threshold, which we use to set upper limits on SME coefficients.

2. Sidereal Time Conversion

Sidereal time describes the Earth's rotation relative to a fixed LV background. If LV exists as predicted by the SME, it will appear as sidereal modulations in the neutrino event rate [4]. In this analysis we

will only consider Local sidereal phase (LSP), which is a float value between 0 and 1 that describes a fraction of the sidereal day.

The timestamp for each data beam spill is recorded by the GPS in Universal Coordinated Time (UTC) and saved in UNIX time format. We convert UNIX to local mean sidereal time. The resulting sidereal time is expressed in sidereal hours, which is normalized by 24 to obtain LSP. A new TTree branch is created which contains each associated data spill time in the LSP format (implementation details described in listing 1). We will use simulated Monte-Carlo (MC) spills to construct null experiments and increase statistics. Each MC spill is assigned a LSP value. The LSP value is randomly drawn from the data spills distribution while the beam exposure remains static. Randomly assigning spills LSP values removes the effect of any physical modulation in the simulated MC spills distribution, as shown in figure 1.

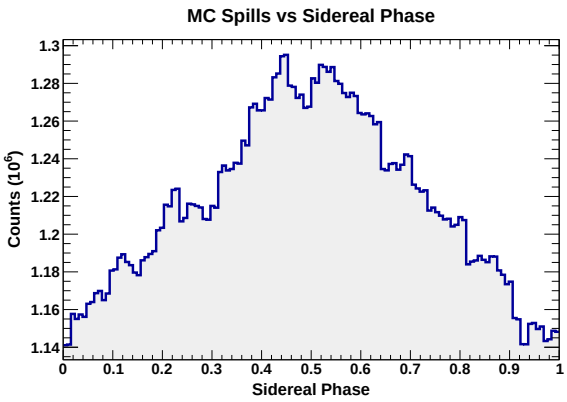

Figure 1. Distribution of simulated spills.

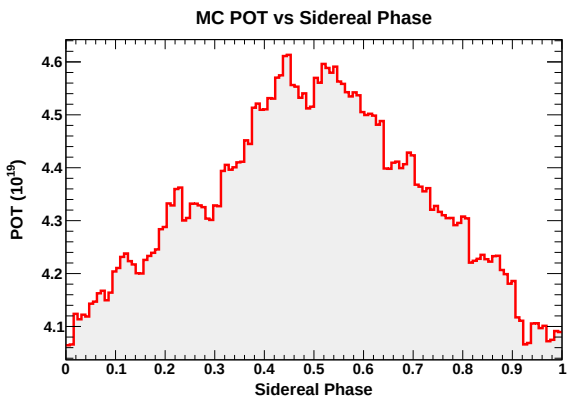

Figure 2. Distribution of simulated protons delivered to the NuMI target as a function of LSP.

We extract the measured protons-on-target from within each data spill, associated with a certain LSP. The total MC POT exposure is weighted down to match the data. A distribution of MC POT as a function of LSP is shown in figure 2. The POT will be used later to normalize the event-rate distribution.

3. Simulated Universes

A simulated experiment is a full measured or MC dataset with randomized LSP values for each selected spill and a fixed total exposure. We use the data interactions to simulate thousands universes with zero sidereal modulation to determine a signal threshold. We use the MC interactions to simulate universes with a certain amount of LV effects to set constraints on each of the SME coefficients. The two methods of simulating universes are very similar, differing in the way we generate their respective event rates.

We simulate thousands of random null experiments to constrain the value of each SME coefficient. Each random experiment follows an identical procedure which is defined as follows.

3.1. Event and Spill Definitions

Each MC event has several characteristics. Event data is organized by a collection of data records which is used across the analysis. A description of each event characteristic stored in the data records is listed in table 3. The collection of records is populated by one immutable record for each unique event. Events are unique if they have differing `EventID`'s,

$$\texttt{EventID} = \texttt{run} \times 10^{12} + \texttt{subrun} \times 10^{9} + \texttt{cycle} \times 10^{6} + \texttt{event}$$

Since selected events only occur during beam runs, events are assigned the LSP timestamp of their corresponding spill. Each unique logical spill is characterized by a `SpillID`,

$$\texttt{SpillID} = \texttt{run} \times 10^{12} + \texttt{subrun} \times 10^{9} + \texttt{event}$$

Thus, events from the same spill have identical run, subrun, and event identifiers, while they may differ in cycle number.

3.2. Constructing a Data Event Rate

Since sidereal modulation could exist in the measured data, we must remove any possible LV effects in order to establish a signal threshold. To generate a null event rate, we randomize the LSP of each data spill and its corresponding events by sampling random LSP values from the data POT vs LSP distribution. Next, we fill the POT and events of a given selection into separate distributions as a function of LSP. Dividing the events vs LSP histogram by POT vs LSP yields an event rate as a function of LSP with zero sidereal modulation, as shown in fig 3.

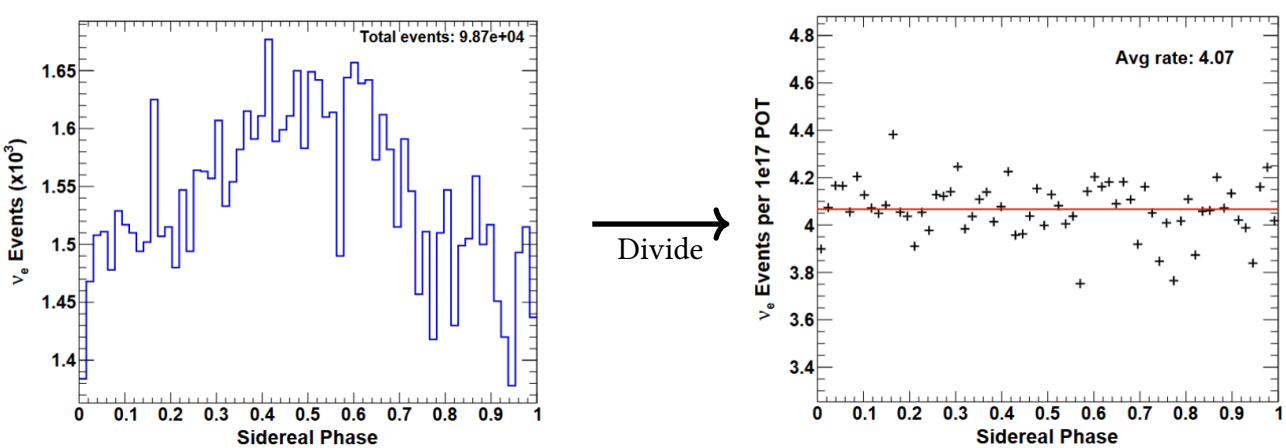

Figure 3. Division of simulated events (left) by POT vs LSP to yield an event rate (right). The average event rate is indicated in red.

3.3. Constructing an MC Event Rate

In the MC sample, a single `SpillID` its associated events may correspond to many MC spills. We therefore define a logical spill as the collection of all MC spills sharing the same `SpillID`. The total POT associated with each logical spill is defined as the sum of the POT from all TTree entries sharing the same `SpillID`.

For each simulated experiment, the LSP value of each logical spill is randomized by drawing from the data POT vs LSP distribution. This LSP value is assigned to the logical spill and all simulated interactions with the corresponding `SpillID`. We divide the total POT from each logical spill by the number of spills within it to obtain an average amount of POT per spill. We fill this average POT into the corresponding randomized LSP bin in a new MC POT histogram. The average multiplicity of spills within a logical spill is listed in table 1.

For each logical spill, we count the number of events of the given selection. Next, we divide this quantity by the number of spills in the logical spill. Thus, this division yields an average number of events per spill. We input this average into a Poisson draw to model spill-to-spill fluctuations. The output of the Poisson draw is the number of events we sample from the corresponding logical spill. If the Poisson draw returns a quantity that exceeds the total number of events in the logical spill, we sample all available events. We fill each sampled events into an events vs LSP histogram, as shown in figure 3 (left). To remove the effects of time-dependent beam intensity, the simulated events histogram is divided by the simulated POT versus LSP histogram. This yields a normalized MC event rate distribution (events per unit POT) that is uniform in the absence of sidereal modulation, with Gaussian statistical fluctuations from the random LSP assignment, as shown in figure 3.

Quantity	ν_μ	ν_e
Data		
Selected Events	6,584,371	20,144
Number of Spills	59,987,695	
Data POT	2.417×10^{21}	
Events / POT	2.724×10^{-15}	8.333×10^{-18}
Monte Carlo		
Selected Events	13,977,421	207,676
Number of Spills	155,662,491	
Number of Logical Spills		
MC POT	5.545×10^{21}	
Events / POT	2.521×10^{-15}	3.745×10^{-17}

Table 1. Total quantity of events, spills, and POT for the ν_e and ν_μ data and MC samples.

3.4. The Fast Fourier Transform

To quantify the amount of sidereal modulation present, we apply the Fast Fourier Transform (FFT) to the simulated event rate distribution using the `SciPy` Python library. The FFT operates on discrete distributions with outputs in the frequency domain in which a set of complex numbers encode both the amplitude and phase of the signal components at each integer multiple of the sidereal frequency [5]. For instance, applying the FFT to a histogram binned into 128 LSP intervals yields 128 complex coefficients. Each coefficient corresponds to a harmonic of the fundamental sidereal frequency, $\omega_\oplus$ with bin index i, representing the ith multiple of $\omega_\oplus$. To quantify the strength or power of a possible modulation at each frequency bin, we compute the magnitude of each complex coefficient using the quadratic sum of its real (C_i) and imaginary (S_i) components:

$$P_i = \sqrt{S_i^2 + C_i^2} \tag{1}$$

Here, P_i represents the signal power in the ith frequency bin, which is used to evaluate the presence of modulations at corresponding sidereal harmonics. The SME predicts power at integer harmonics of $\omega_\oplus$, thus we restrict our analysis to the first four harmonic bins [4].

4. Establishing a Signal Threshold

Next, we simulate 10^4 random experiments, each with identical beam exposure to the real dataset, as described in section 3.2. For each simulated experiment we record the FFT power at the first four sidereal harmonics which are stored in a new histogram. Repeating this process across all 10^4 simulated experiments results in a distribution of expected FFT powers under the null hypothesis. Next, we use a Rayleigh distribution to fit the distribution. The expression describing the Rayleigh fit is shown below:

$$f(x) = \frac{x}{\sigma^2} \exp(-\frac{x^2}{2\sigma^2}) \tag{2}$$

To define our detection threshold, we adopt a 3σ criterion. The threshold power is defined as the value for which 99.7% of the simulated experiments have smaller FFT powers. The lack of sidereal modulation in the simulated neutrino event rate is validated by the quality of the fit to the distribution, as shown in figure 4.

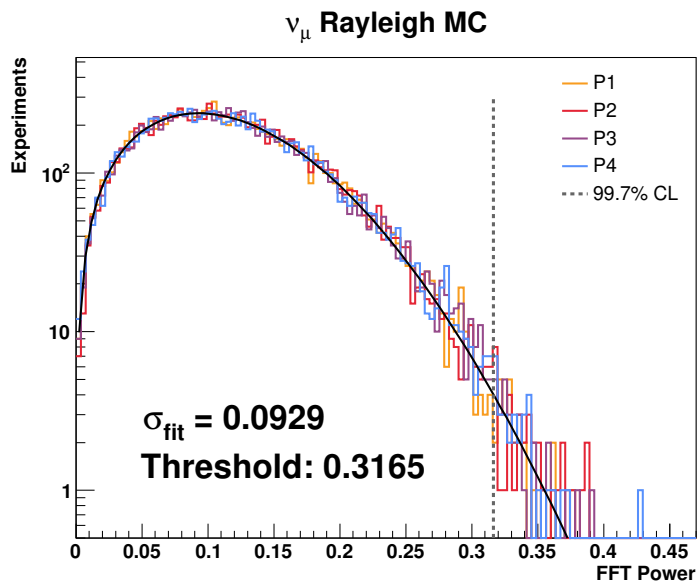

Figure 4. Spectrum of possible FFT power values calculated on independent simulated experiments in the ν_μ channel. The vertical line indicates the 99.7% threshold.

5. Constraining SME Coefficients

In the absence of a statistically significant sidereal modulation in the observed neutrino event rate, we place constraints on Lorentz violating coefficients in the SME. The analysis considers the nine relevant coefficients in $(a_L)^\mu$ and $(c_L)^{\mu\nu}$. We assume that only one coefficient is nonzero at a time, setting all others to zero.

5.1. Modeling LV Driven Oscillations

For short-baseline experiments, the SME treats LV induced oscillations perturbatively. We use the first order LV driven oscillation probability,

$$P_{\mathrm{LV}}(E, L, \omega_\oplus T_\oplus) = 1 - \frac{L^2}{(\hbar c)^2} \left|\mathcal{H}_{\mathrm{eff}}\right|^2, \tag{3}$$

where $\mathcal{H}_{\mathrm{eff}}$ is the effective Hamiltonian defined in [4]. We use the true neutrino energy and true decay position from the MC, while considering the spread in pion decay locations along the beamline. Beam-geometry and cross-section weights provided by the Monte Carlo are applied to each event. Figure 5 shows the resulting spectrum for MC events in energy and baseline.

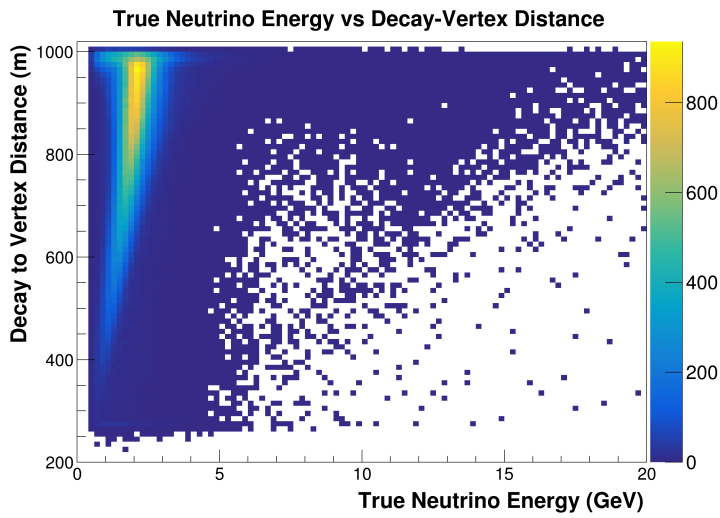

Figure 5. Distribution of true neutrino events as a function of true energy and decay-to-vertex distance. Courtesy of Anna Cooleybeck of the UW–Madison neutrino group.

5.2. LV Event Rate

For a given SME coefficient value, we create event rate histograms with no sidereal modulation for both the ν_e and ν_μ oscillation channels, as described in section 3.3. For each MC event, the LV oscillation probability P_{LV} is computed using equation 3. Each event's oscillation weight is multiplied by the probability that the event is reconstructed and passes selection criteria. This probability is defined as the detector efficiency $\epsilon(E)$, which depends on the interaction mode, oscillation channel, and true neutrino energy. To preserve total probability, a weight of ϵP_{LV} is filled to the histogram corresponding to the oscillated flavor, while a weight of $\epsilon(1 - P_{\mathrm{LV}})$ is filled to the histogram corresponding to the original flavor. Figure 6 shows example appearance and disappearance distributions for the ν_e channel with a single non zero SME coefficient.

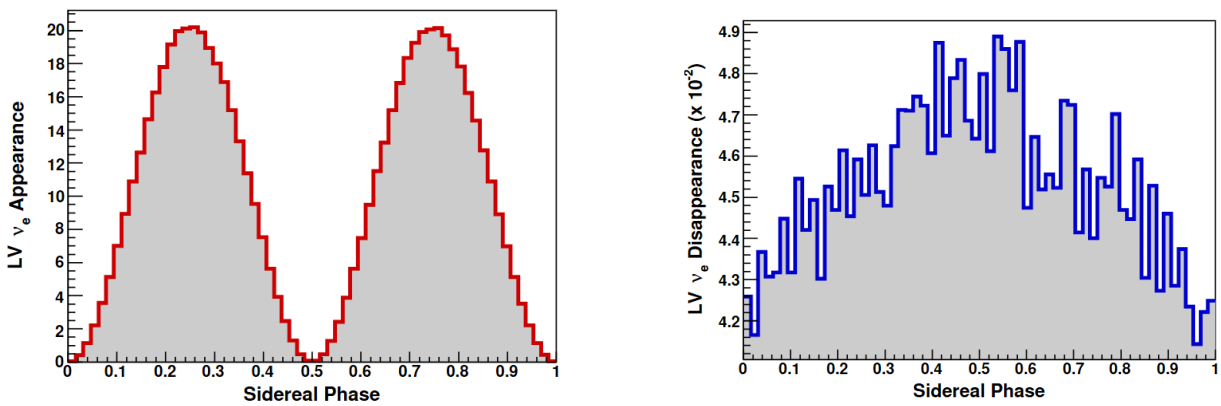

Figure 6. Example LV-induced appearance (left) and disappearance (right) event rate distributions for the ν_e channel with a single nonzero SME coefficient.

5.3. Statistical Procedure and Limit Definition

For each SME coefficient, the coefficient value is scanned logarithmically over a predefined range. At each value, 1000 pseudo experiments are performed. For each trial, the LV modulated event rate is constructed the FFT spectrum is recorded for both channels. A trial is classified as signal if the FFT power in the first four harmonic bins between both oscillation channels exceeds the Rayleigh threshold. For a given coefficient value, the fraction of trials exceeding threshold is computed. The coefficient value is increased until at least 95% of trials exceed the Rayleigh threshold. The smallest coefficient value satisfying this criterion is set as the 3σ limit for that coefficient. The resulting limits for all coefficients are summarized in table 2.

SME Coefficient	$\nu_\mu \to \nu_e$	$\nu_e \to \nu_e$	$\nu_\mu \to \nu_\mu$
$(a_L)^X$	TBD	TBD	TBD
$(a_L)^Y$	TBD	TBD	TBD
$(a_L)^Z$	TBD	TBD	TBD
$(c_L)^{TX}$	TBD	TBD	TBD
$(c_L)^{TY}$	TBD	TBD	TBD
$(c_L)^{TZ}$	TBD	TBD	TBD
$(c_L)^{XY}$	TBD	TBD	TBD
$(c_L)^{XZ}$	TBD	TBD	TBD
$(c_L)^{YZ}$	TBD	TBD	TBD

Table 2. Calculated 3σ limits on coefficients predicted by the SME.

Figure 7 shows the ν_e event rate for the $(a_L)^X$ coefficient set to its corresponding limit value, along with the simulated detector excess relative to the nominal event rate. The coefficient limit calculation procedure is summarized in figure 8.

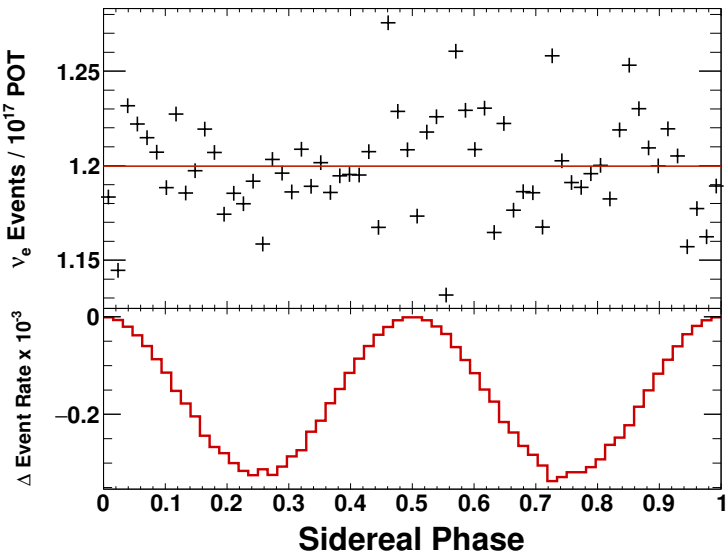

Figure 7. LV-modulated ν_e event rate at the coefficient value corresponding to the 3σ sensitivity limit (top), and the excess relative to the nominal event rate (bottom).

6. Acknowledgments

We greatly appreciate the contributions to this analysis by the UW-Madison ν-group, NOνA collaboration, and FermiLab staff.

7. Appendix

7.1. Sidereal Time Conversion Algorithm

To perform the conversion between UNIX and LSP timestamps, we use the `AstroPy` Python library as follows:

- Read ROOT TTrees from a `.root` file.
- Extract the `spillTimeSec` and `spillPOT` branches.
- For each timestamp, we use `AstroPy` to compute the local sidereal time (LST) at Fermilab's location [1].
- Save the new sidereal time branch into a ROOT TTree.

Listing 1. Sidereal Conversion using AstroPy

```
from astropy.time import Time
from astropy.coordinates import EarthLocation

out_tree = ROOT.TTree("CombinedSpills", "Combined␣spill␣data␣with␣LST") # Initialize Output TTree

fermilab = EarthLocation(lat=41.8406, lon=-88.2616, height=228)

spill_time = tree["spillTimeSec"].array(library="numpy") # Getting UNIX Time Branch
pot = tree["spillPOT"].array(library="numpy") # Getting POT Branch

t_astropy = Time(spill_time, format="unix", scale="utc", location=fermilab)
lst = t_astropy.sidereal_time("mean").hour # Calculate mean LST

lst_mean[0] = lst[i]
out_tree.Fill() # Fill entry to output TTree branch
```

7.2. Data Record Characteristics

`Local Sidereal Phase`	A float value between 0 and 1 which describes the phase within a sidereal day.
`EventID`	A long integer which characterizes unique events.
`Logical SpillID`	A long integer which characterizes unique logical spills.
`Logical Protons on Target (POT)`	A long integer which describes the total POT in a logical spill.
`True Energy`	A float value which is output from the simulation describing the energy of the interaction.
`True Baseline`	A float value which is output from the simulation describing the distance, in meters, from the mother decay in the decay pipe to the interaction vertex.
`Selection Number`	A integer (0 or 1) describing reconstructed neutrino flavor.
`PDG Code`	Corresponding flavor of the neutrino output by the simulation.
`Interaction Type`	A string which describes the mode of interaction with ions in the detection.
`Detection Efficiency`	The fraction of all simulated events which are reconstructed and selected for the corresponding neutrino flavor and interaction mode.

Table 3. All event characteristics stored in each data record.

7.3.

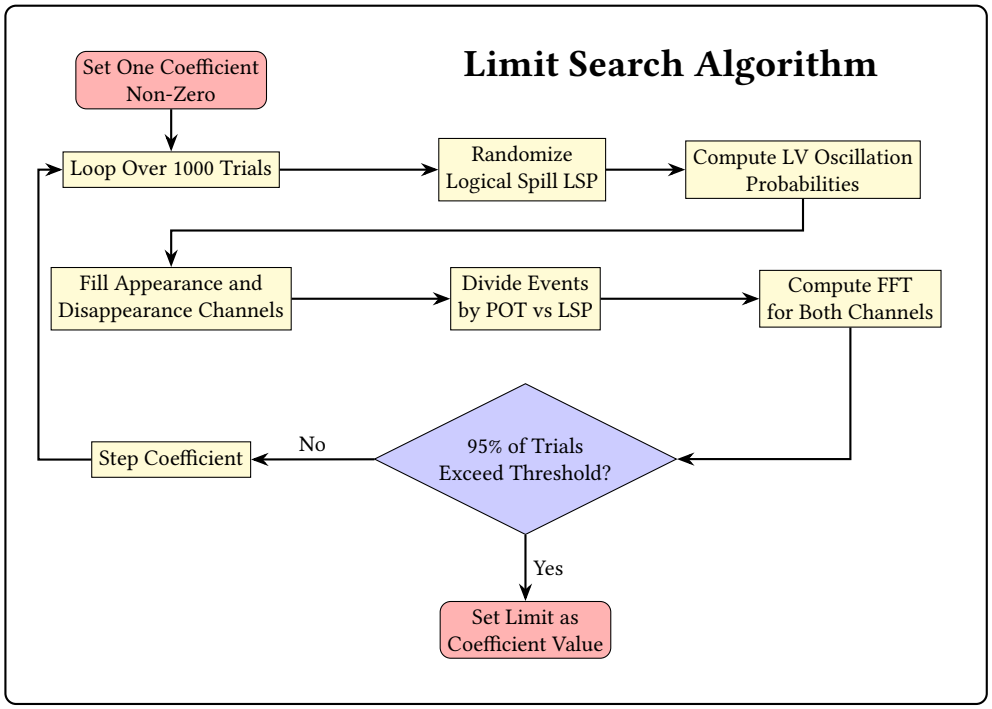

Figure 8. Flowchart of the limit calculation procedure using randomized simulated trials.

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